Dear Editors:

We would like to submit the enclosed manuscript entitled “Too Much of a Good Thing? How the Depth of Generative Artificial Intelligence Adoption Shapes Youth Employment in Chinese Listed Firms” for consideration in the Special Issue “Systems Approaches to Generative AI: Workforce Development, Organisational Learning, and Economic Transformation” of Systems. The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

Whether generative artificial intelligence destroys or creates entry-level employment is one of the most consequential open questions in labour economics, and the firm-level evidence is flatly contradictory: some studies find sharp displacement of young workers, others find nothing. This manuscript argues that the contradiction is a functional-form artefact and shows that it is. Using 27,320 firm-year observations for 3,400 Chinese A-share listed firms over 2016–2025 and a text-based measure of adoption depth, we find that the relationship is not monotone but inverted U-shaped.

The contributions of the paper are as follows.

(1) It resolves a live empirical contradiction with a functional form. A linear specification with firm and year fixed effects and a full control vector finds nothing at all (−0.019, $p = 0.80$). Adding the squared term reveals a strongly significant inverted U (2.952 on depth, −0.692 on its square). Studies that estimate a monotone relationship are averaging across a rising and a falling arm, which is why their signs disagree.

(2) It tests the shape properly rather than reading it off a quadratic coefficient. The Lind–Mehlum exact test rejects monotonicity ($t = 13.91$, $p < 0.001$); the extreme point, 2.132, has a Fieller interval of [2.03, 2.24] lying wholly inside the observed range, with 42.9% of post-shock observations already beyond it; a two-lines test finds significant slopes of opposite sign on either side; and a non-parametric binned fit traces the same curve.

(3) It explains why the curve bends. The augmentation of entry-level work is increasing and concave, the automation of the entry-level task bundle increasing and convex, the youth share loads positively on the first and negatively on the second, and once both are controlled for no residual curvature remains. Moderate adoption raises the share of employees aged 30 or below by up to 3.15 percentage points; the deepest observed adoption leaves the firm 1.84 points below where it started.

(4) It shows that the turning point is an organisational variable, not a technological constant. Organisational learning capability delays it by 0.51 units of depth per standard deviation and disclosed AI governance by 0.67, while labour cost pressure brings it forward by 0.56. The managerial question becomes how deep a firm can go before the entry-level role has to be rebuilt, and how much further organisational design can push that frontier.

We believe the manuscript speaks directly to the aims of the Special Issue. It treats generative AI adoption as an organisational rather than a purely technological event, identifies the learning and governance mechanisms that determine where a firm can operate on the curve, and converts them into levers that firms and policymakers can actually pull. Identification rests on two shift-share instruments and a peer-diffusion instrument that pass the Kleibergen–Paap and overidentification tests, and every number in every table and figure is produced by the released estimation code. We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to the address below.

Thanks very much for your attention to our paper.

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Very sincerely yours,
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